Activity No. 13	
Working with Files and Directories in Linux	
Course Code: CPE 201A	**Program: BSCPE**
Course Title: COMPUTER SYSTEM ADMINISTRATION AND TROUBLESHOOTING	**Date Performed:**
Section: CPE11S4	**Date Submitted: October 22, 2026**
Name: Altea Kara A. Blaza	**Instructor: Engr. Noel Jason Lusung**

1. Objective/s:

This activity aims to navigate and manage files and directories using command line interface.

2. Intended Learning Outcome/s:

The students should be able to:

2.1 Demonstrate how to navigate home and system directories
2.2 Demonstrate how to list files and directories.
2.3 Demonstrate how to use globbing to manipulate files and directories.
2.4 Demonstrate how to create, move and delete files and directories.

3. Discussion:

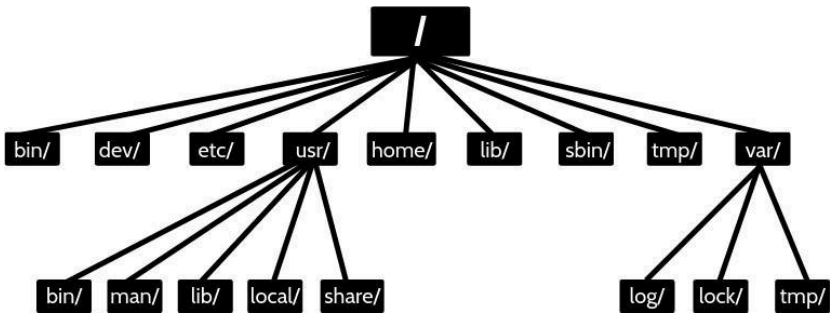
Files and Directories

The Linux filesystem is similar to other operating system's filesystems in that it contains *files* and *directories*. Files contain data such as human-readable text, executable programs, or binary data that is used by the computer.

Directories are used to create organization within the filesystem. Directories can contain files and other directories.

Directory Structure

On a Windows system, the *top level* of the directory structure is called ***My Computer.*** The Linux directory structure, called a ***filesystem***, also has a top level called the ***root directory*** (symbolized by the **slash /character**).



Home Directory

On most Linux distributions there is a directory called **home** under the root /directory. Under this **/home** directory there is a directory for each user on the system. When a user opens a shell, they should

automatically be placed in their home directory. The user has the full control to create and delete additional files and directories in their home directory. Most other directories in a Linux filesystem are protected with *file permissions*. The home directory has a special symbol used to represent it, the **tilde ~ character**.

The directory name is the same as the name of the user. So, a user named sysadmin would have a *home directory* called /home/sysadmin:

File and Directory Names

File and directory names in Linux can contain lower case and upper case letters, numbers, spaces and special characters. However, since many special characters have a special meaning in the Linux shell, it is good practice to not use spaces or special characters when naming files or directories. Spaces, for example, need the *escape character* \ to be entered correctly

Paths

A path is a list of directories separated by the / character. There are two types of paths: *absolute* and *relative*. For example, /home/sysadmin is a path to the home directory

A. Absolute Paths

Absolute paths allow the user to specify the exact location of a directory. Absolute paths always start at the root directory, and therefore it always begins with the / character.

B. Relative Paths

A relative path gives directions to a file relative to the current location in the filesystem. The user must currently be in a directory that contains objects in the path. Relative paths start with the name of a directory.

Special Relative Paths

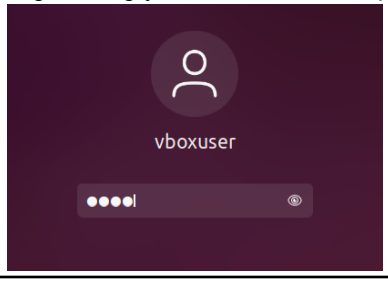
The single period . character always represents the current directory.
Two periods .. characters always represents one directory higher relative to the current directory, sometimes referred to as the parent directory.

4. Resources:

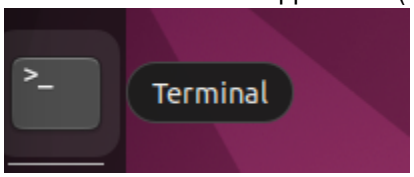
Personal Computer with installed Virtual Box
Ubuntu Server or Desktop virtual machine

5. Procedure:

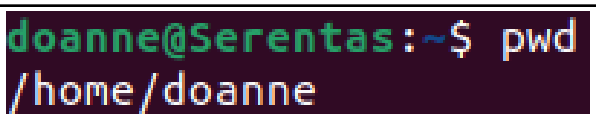
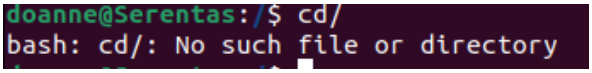
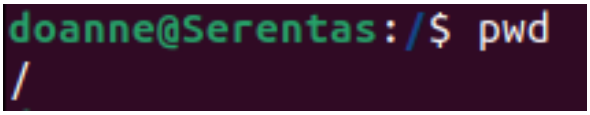
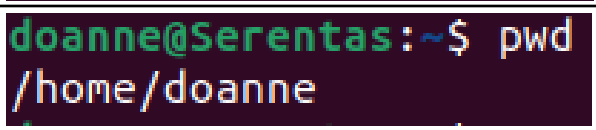
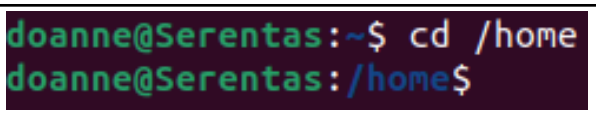
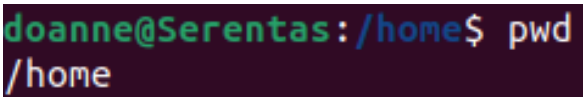
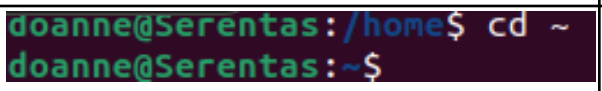
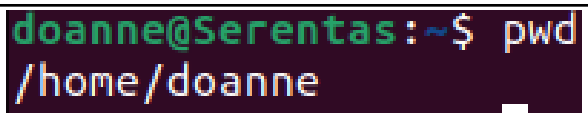
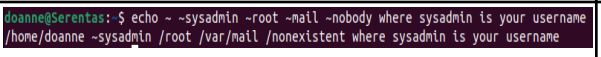
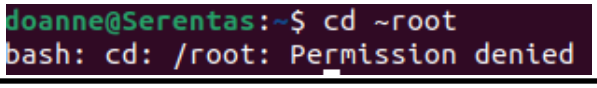
1. Login using your username and password.

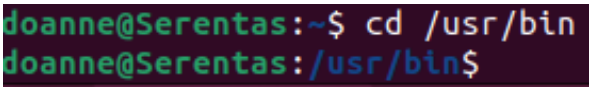
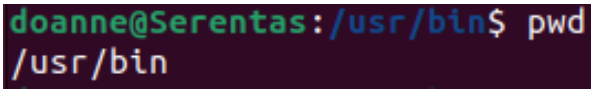
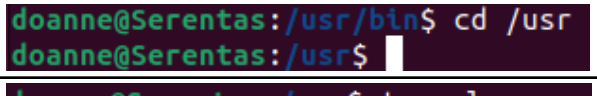
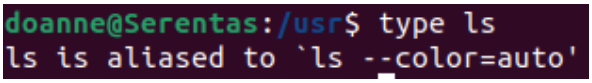
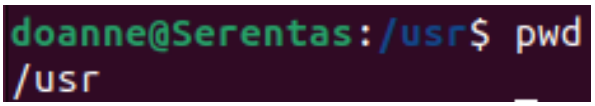
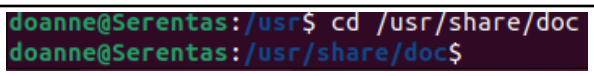
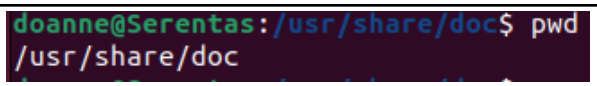
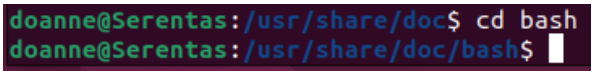
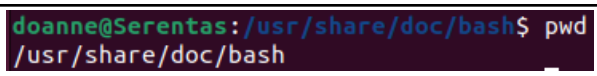
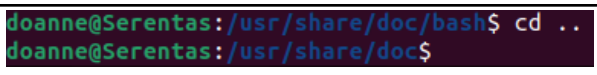
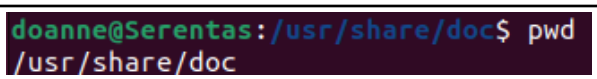
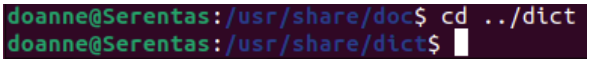


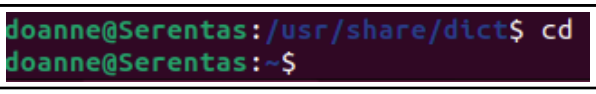
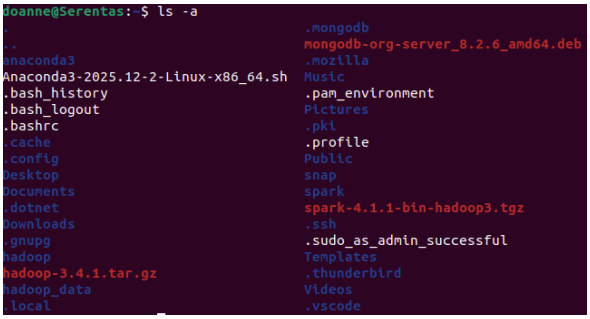
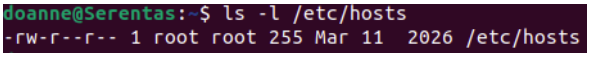
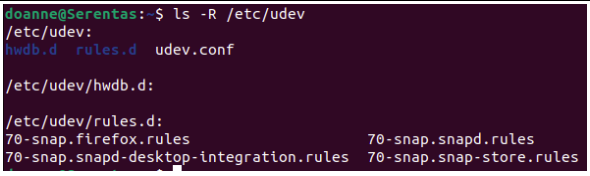
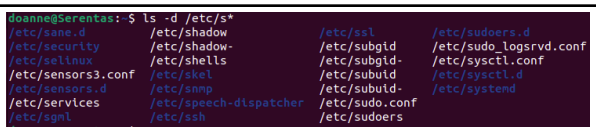
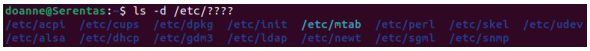
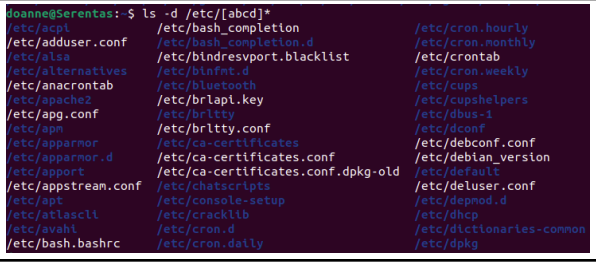
2. Use terminal emulator application (if you are using desktop version)



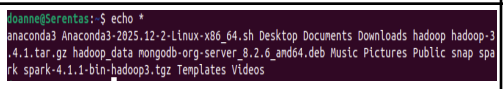
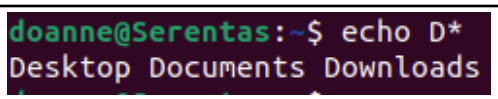
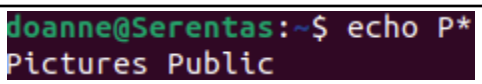
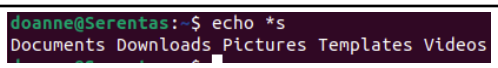
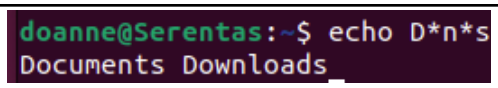
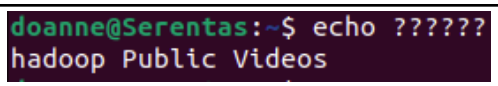
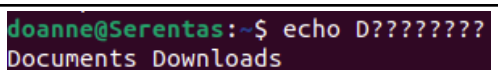
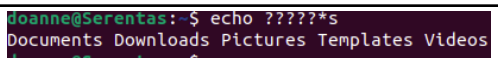
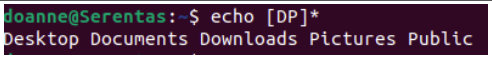
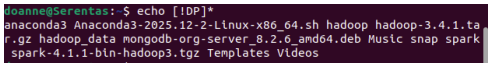
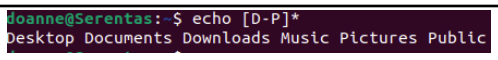
3. Execute the following commands to navigate files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

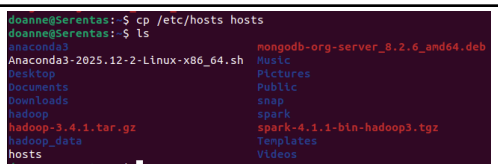
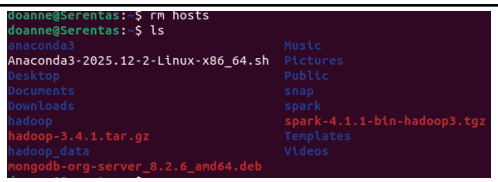
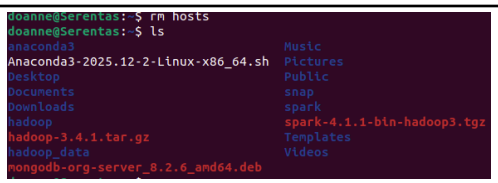
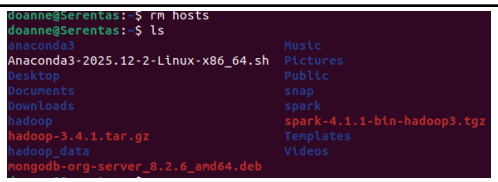
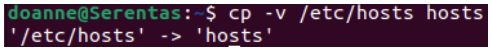
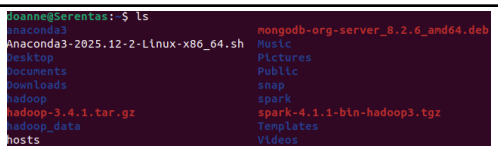
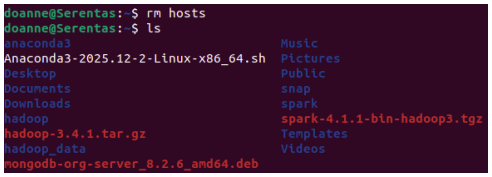
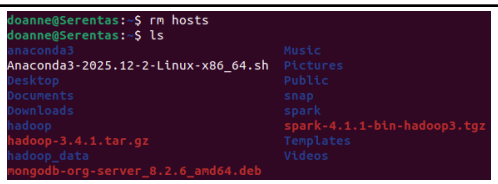
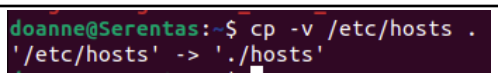
Command	Screenshot	Explanation
1. pwd		Displays the current working directory.
2. cd /		Moves to the root directory of the Linux system.
3. pwd		Confirms that the current directory is now the root directory (/).
4. cd		Returns to my home directory.
5. pwd		Displays my current home directory.
6. cd /home		Moves to the /home directory, where user directories are stored.
7. pwd		Shows the current directory, which should be /home.
8. cd ~		Returns to my home directory using the ~ symbol.
9. pwd		Confirms the location of my home directory.
10. echo ~ ~sysadmin ~root ~mail ~nobody where sysadmin is your username		Displays the home directory paths represented by the tilde shortcuts for the specified users.
11. cd ~root		Attempts to move to the root user's home

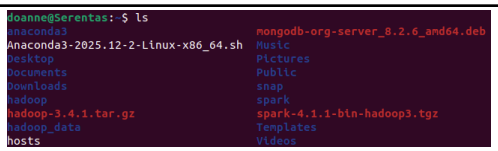
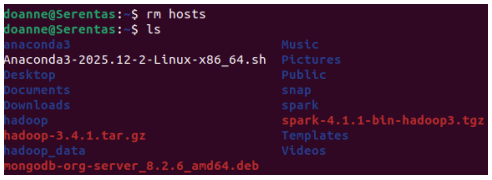
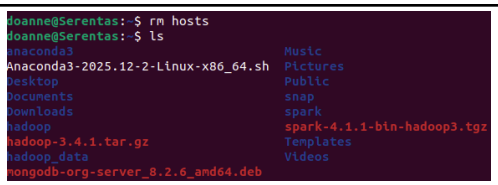
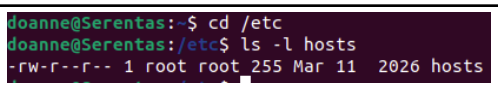
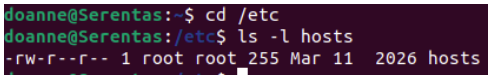
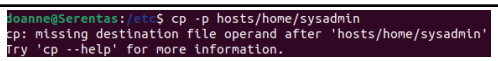
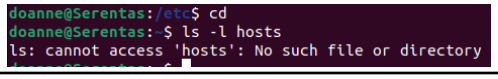
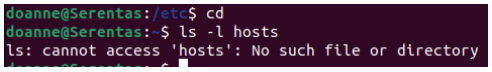
		directory. Access may be denied depending on permissions.
12. cd /usr/bin		Moves to the directory containing many executable programs and commands.
13. pwd		Displays the current directory, which should be `/usr/bin`.
14. cd /usr		Moves to the `/usr` directory.
15. type ls		Identifies how the shell recognizes the `ls` command.
16. pwd		Shows the current working directory, which should be `/usr`.
17. cd /usr/share/doc		Moves to the directory containing documentation for installed software.
18. pwd		Confirms that the current directory is `/usr/share/doc`.
19. cd bash		Moves into the `bash` documentation directory.
20. pwd		Displays the current directory, which should be `/usr/share/doc/bash`.
21. cd ..		Moves one level up to the parent directory.
22. pwd		Shows the current directory after moving one level up.
23. cd ../dict		Attempts to move to the `dict` directory inside the parent directory.
24. pwd		Displays the current location after the previous command.

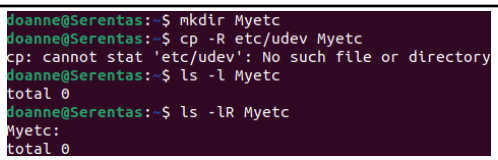
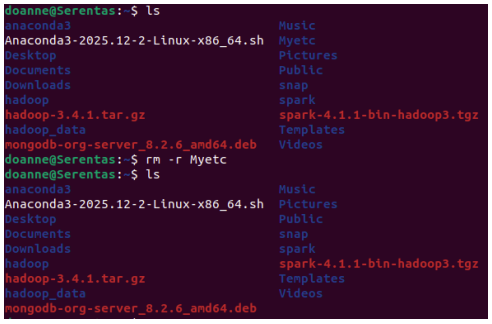
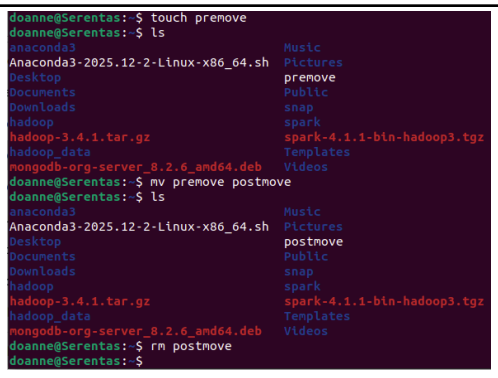
25. cd		Returns to my home directory.
26. ls		Lists the visible files and directories in my home directory.
27. ls -a		Lists all files and directories, including hidden ones.
28. ls -l /etc/hosts		Displays detailed information about the `/etc/hosts` file, such as its permissions, owner, size, and modification date.
29. ls -R /etc/udev		Lists the files and subdirectories inside `/etc/udev`, including those in its subdirectories.
30. ls -d /etc/s*		Displays the names of entries in `/etc` that begin with the letter `s`.
31. ls -d /etc/????		Displays entries in `/etc` whose names contain exactly four characters.
32. ls –d /etc/[abcd]*		Displays entries in `/etc` whose names begin with `a`, `b`, `c`, or `d`.

4. Execute the following commands to manage files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

Command	Screenshot	Explanation
1. echo *		Displays the names of all non-hidden files and directories in the current directory.
2. echo D*		Displays names that begin with the letter D.
3. echo P*		Displays names that begin with the letter P.
4. echo *s		Displays names that end with the letter s.
5. echo D*n*s		Displays names that begin with D, contain n later, and end with s.
6. echo ??????		Displays names that contain exactly six characters.
7. echo D????????		Displays names that begin with D followed by exactly eight characters.
8. echo ?????*s		Displays names with at least five characters before the final s.
9. echo [DP]*		Displays names that begin with either D or P.
10. echo [!DP]*		Displays names that do not begin with D or P.
11. echo [D-P]*		Displays names that begin with a character in the range D through P.
12. echo [!D-P]*		Displays names that do not begin with a character in the range D through P.

13. ls		Lists the files and directories in the current directory.
14. cp /etc/hosts hosts		Copies the `/etc/hosts` file into the current directory and names the copy `hosts`.
15. ls		Checks whether the copied `hosts` file appears in the directory.
16. rm hosts		Deletes the copied `hosts` file.
17. ls		Checks the directory after deleting the file.
18. cp –v /etc/hosts hosts		Copies the file and displays a message describing the copying operation.
19. ls		Lists the directory to check the copied file.
20. rm hosts		Removes the copied `hosts` file.
21. ls		Checks the directory after removing the file.
22. cp –v /etc/hosts .		Copies `/etc/hosts` into the current directory. The dot (`.`)

		represents the current directory.
23. ls		Checks whether the copied file is present.
24. rm hosts		Deletes the copied `hosts` file from the current directory.
25. ls		Lists the remaining files and directories.
26. cd /etc		Moves to the `/etc` directory, which contains system configuration files.
27. ls -l hosts		Displays detailed information about the `hosts` file.
28. cp –p hosts/home/sysadmin		Copies the `hosts` file to `/home/sysadmin` while preserving its original permissions, ownership where permitted, and timestamps.
29. cd		Returns to my home directory.
30. ls –l hosts		Displays detailed information about the `hosts` file in my home directory.
31. rm hosts 32. cp -p /etc/hosts ~ 33. cp hosts newname 34. ls –l hosts newname 35. rm hosts newname		31.Deletes the `hosts` file from my home directory. 32. Copies `/etc/hosts` to my home directory while preserving its file attributes where permitted.

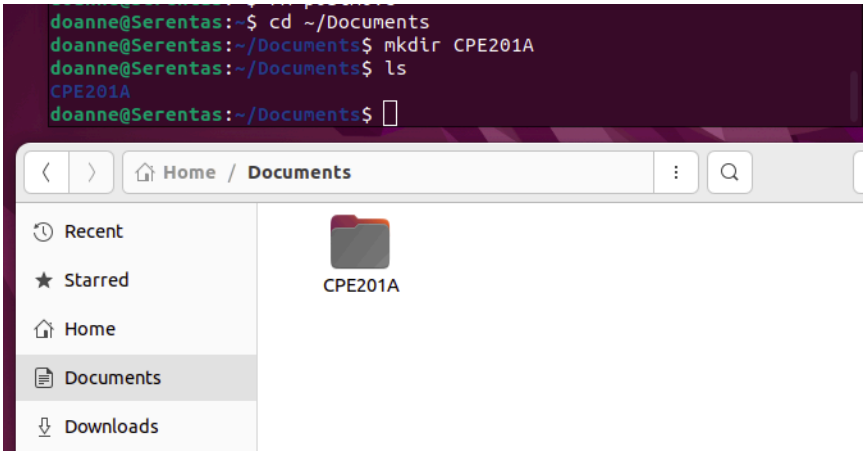
		33. Creates another copy of `hosts` named `newname`. 34. Displays detailed information about both files. 35. Deletes both `hosts` and `newname`.
36. mkdir Myetc 37. cp –R /etc/udev Myetc 38. ls –l Myetc 39. ls –lR Myetc		36. Creates a new directory named `Myetc`. 37. Copies the `udev` directory and its contents recursively into `Myetc`. 38. Lists the contents of `Myetc` with detailed file information. 39. Displays detailed listings of `Myetc` and all its subdirectories recursively.
40. ls 41. rm -r Myetc 42. ls		40. Lists the files and directories in the current directory. 41. Deletes the `Myetc` directory and its contents recursively. 42. Checks the directory after removing `Myetc`.
43. touch premove 44. ls 45. mv premove postmove 46. ls 47. rm postmove		43. Creates an empty file named `premove`, or updates its timestamps if it already exists. 44. Checks whether the `premove` file appears in the directory. 45. Renames the file from `premove` to `postmove`. 46. Checks that the file now has the name `postmove`.

		47. Deletes the `postmove` file.

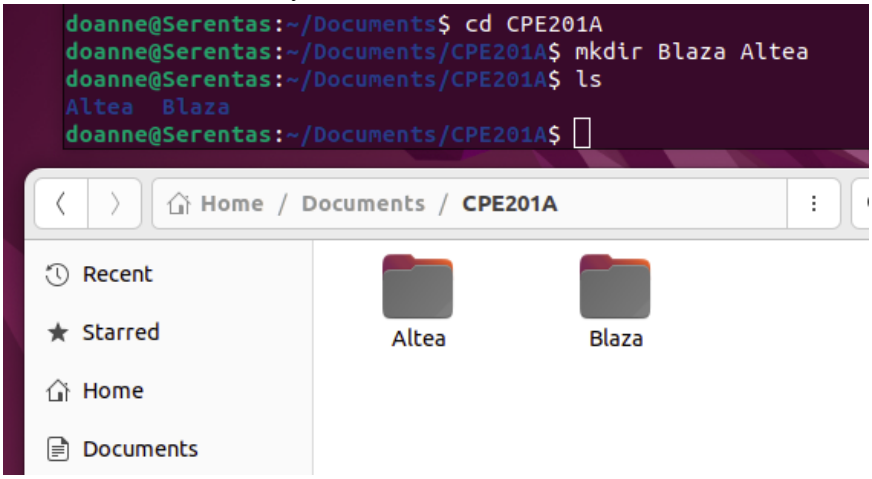
6. Supplementary Activity:

Copy screen shot(s) of the following tasks:

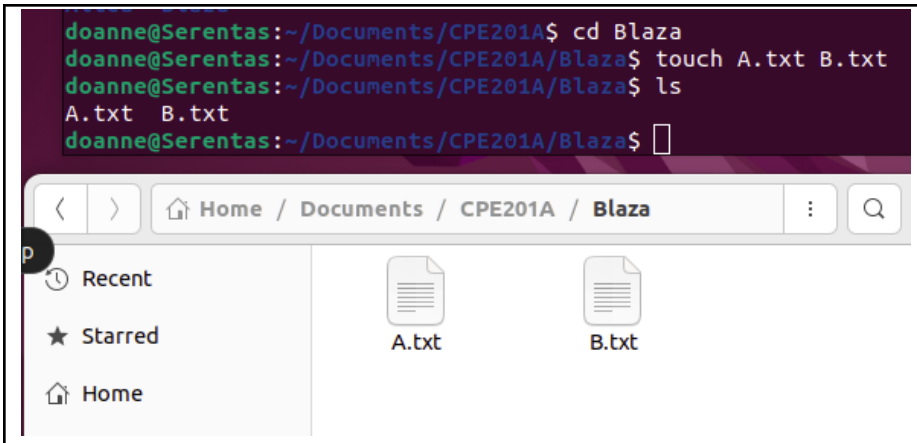
1. Create a directory CPE231 in the Documents directory.



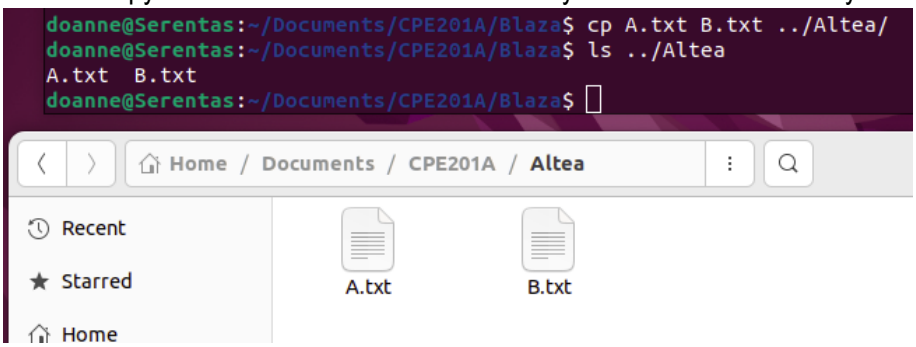
2. Create a directory <lastname> and <firstname> under CPE231 directory.



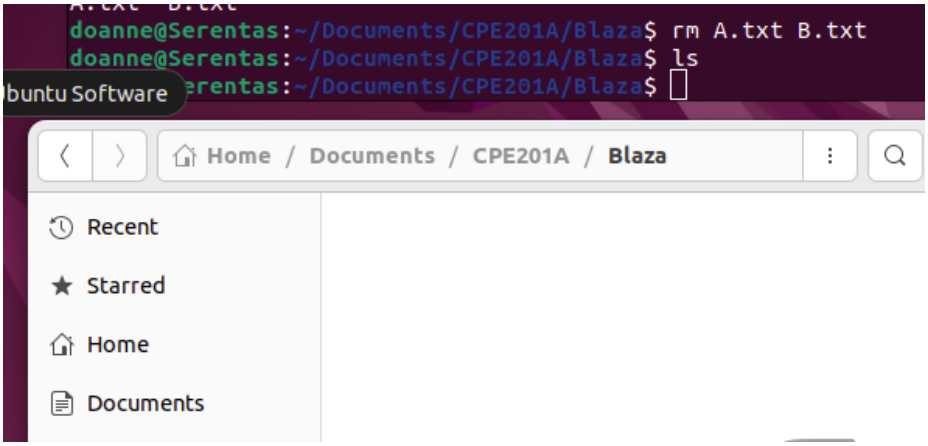
3. Create two text files: (1) A.txt and (2) B.txt in the <lastname> directory.



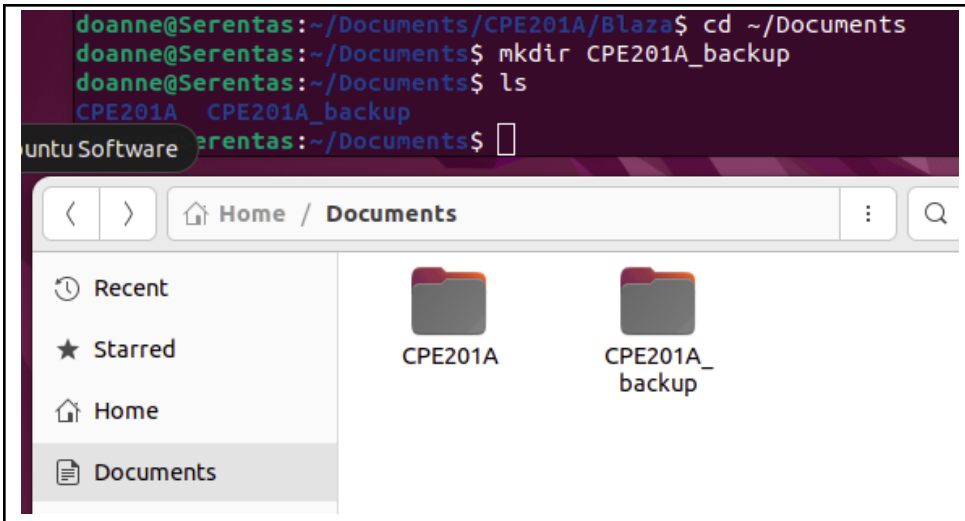
4. Copy the content of <lastname> directory to <firstname> directory.



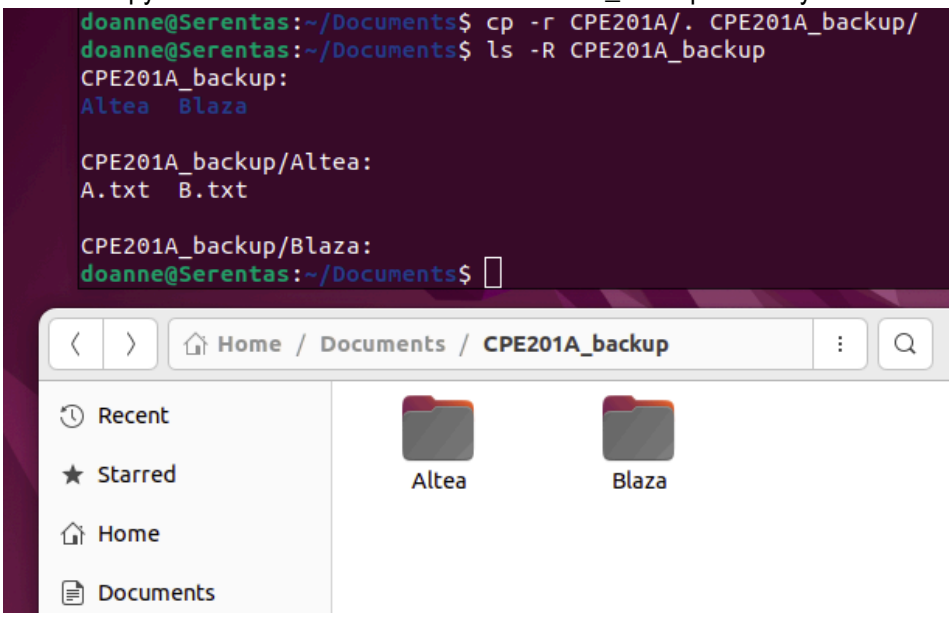
5. Delete the content of <lastname> directory.



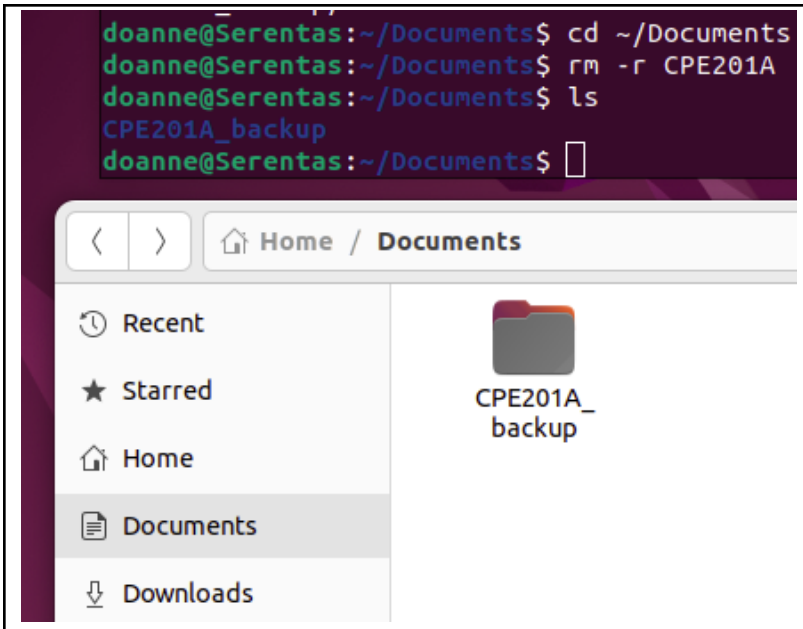
6. Create a directory CPE231_backup in the Documents directory.



7. Copy the content of CPE231 to the CPE231_backup directory.



8. Remove the CPE231 directory.



7. Conclusion:

In this activity, I learned how to navigate the Linux file system and use commands to list, create, copy, move, rename, and delete files and directories. I also learned how wildcards work and how they make searching for files easier. To conclude, this activity helped me become more familiar with using the Linux terminal and understand how to manage files and directories through commands.

8. Assessment (Rubric for Laboratory Performance):